CS & IT ENGINEERING

THEORY OF COMPUTATION

REGULAR EXPRESSION

Lecture No.- 04

By- Venkat sir

Recap of Previous Lecture

Topic ?????

→ Finite Automata ⟹ Regular Expression

→ State elimination method

→ Arden's method

Topics to be Covered

Topic — F. A ⇒ Regular Expression

Topic ??

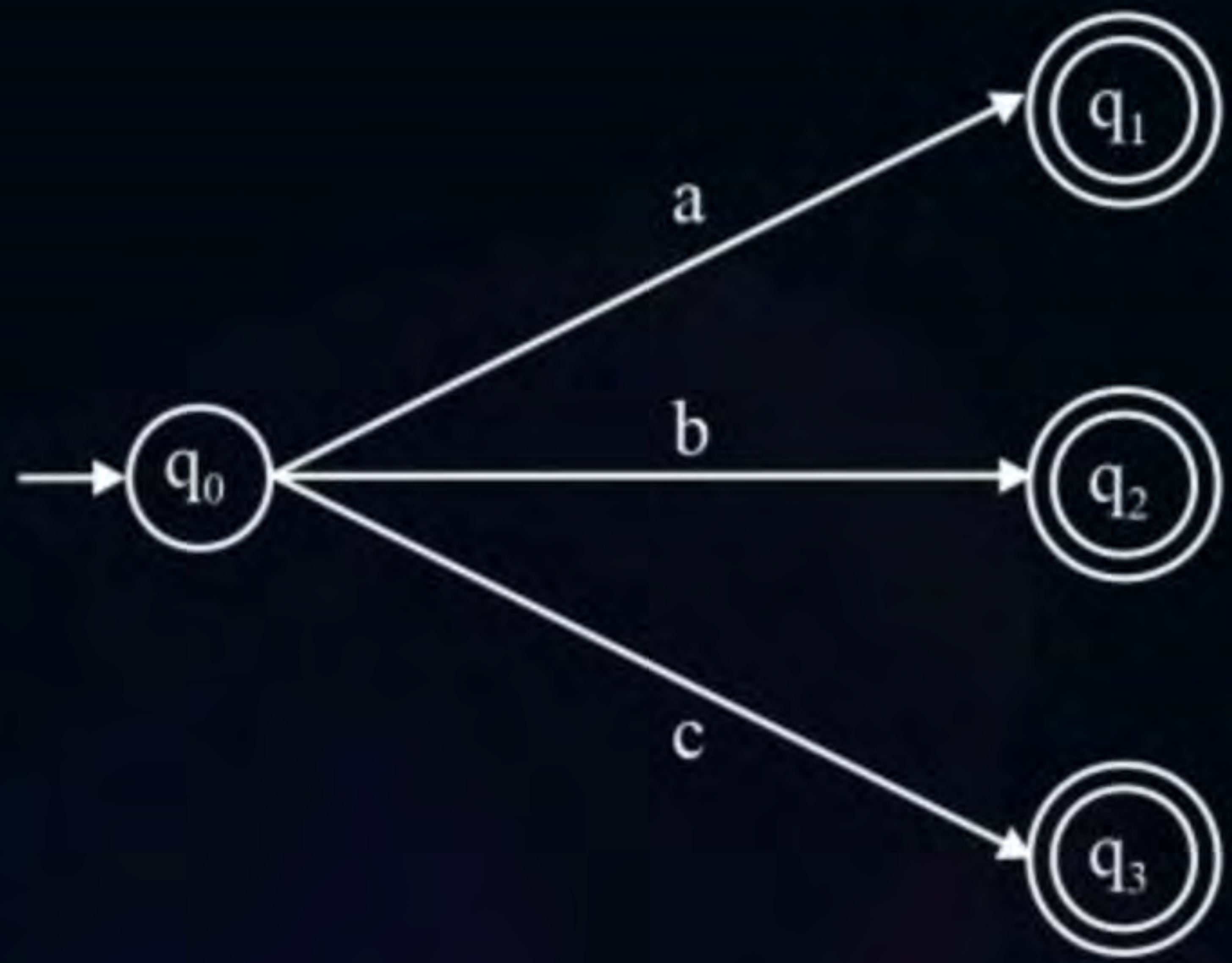
Regular Expression ⇒ Finite Automata

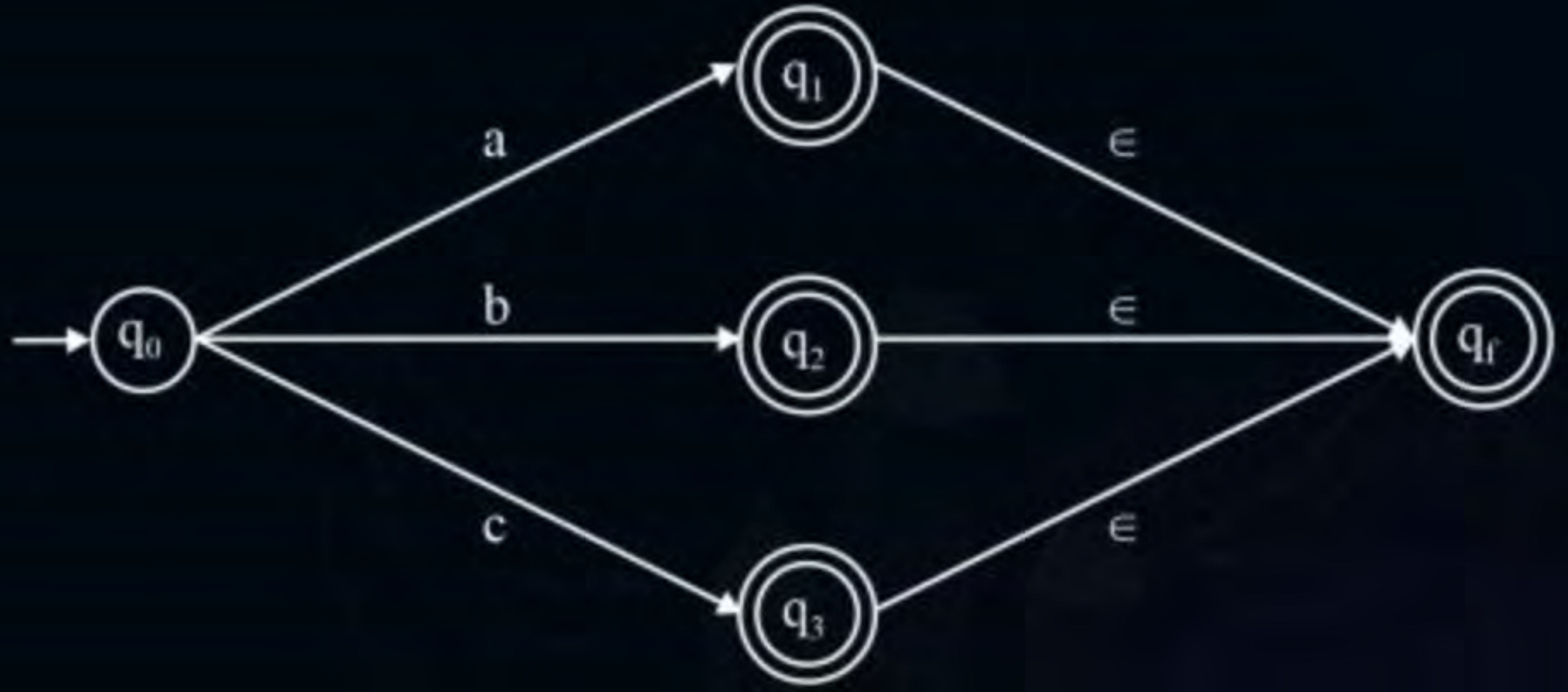
Topic ??

Topic ??

FINITE AUTOMATA TO REGULAR EXPRESSION

1.





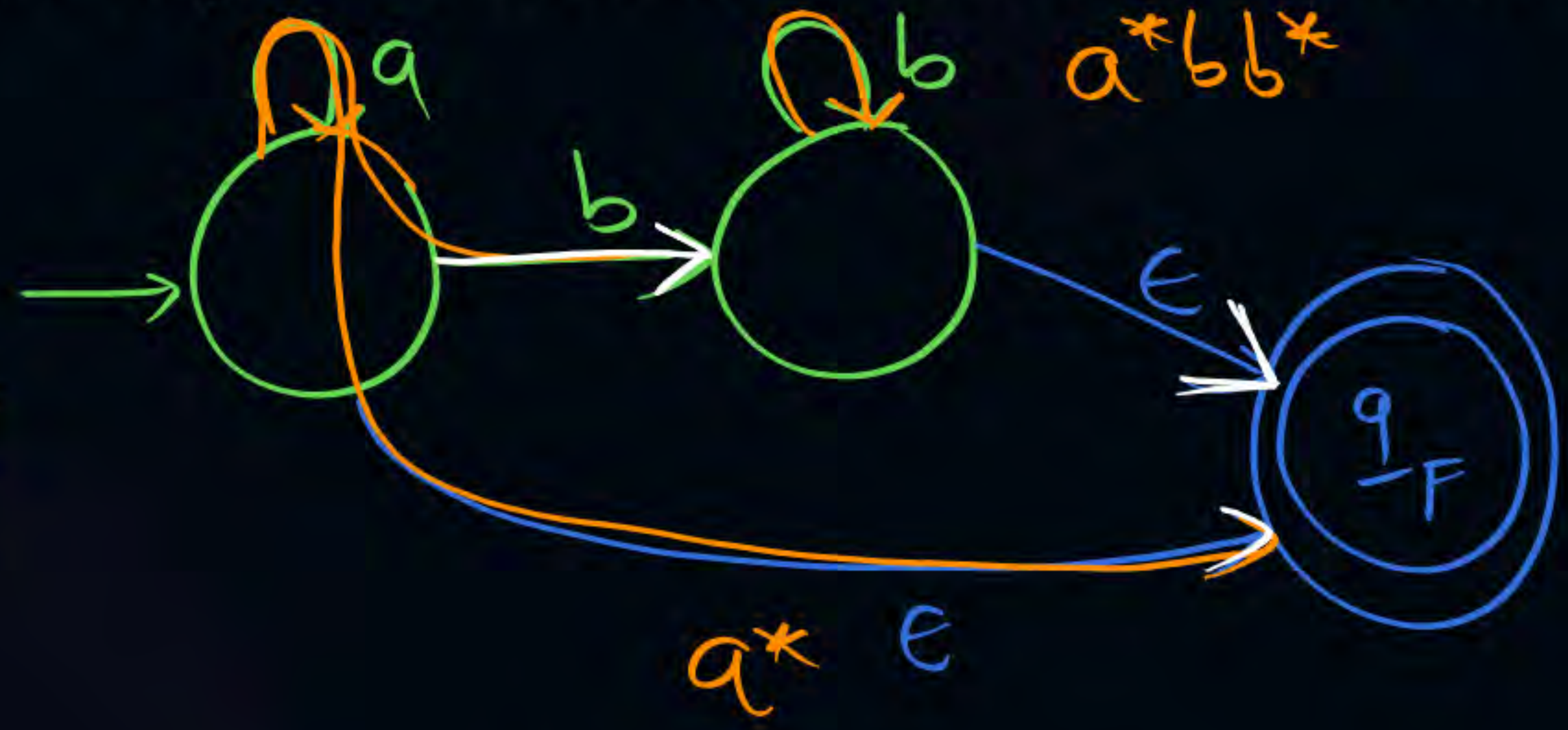
2.

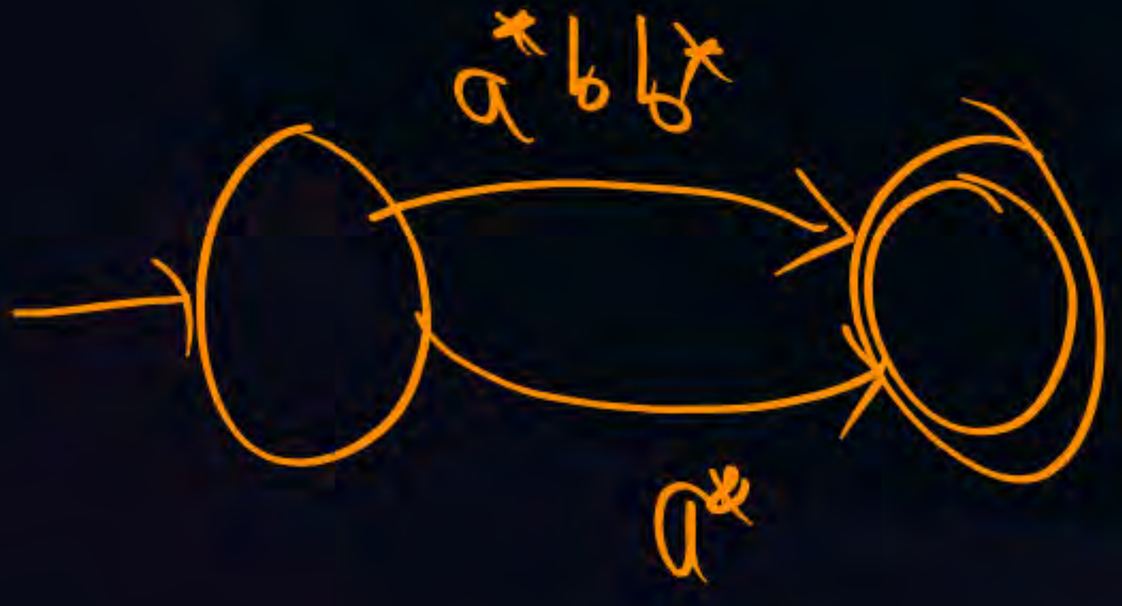
3.

[NAT]

#Q. Construct Regular Expression for the following Finite Automata.

a^*b^*



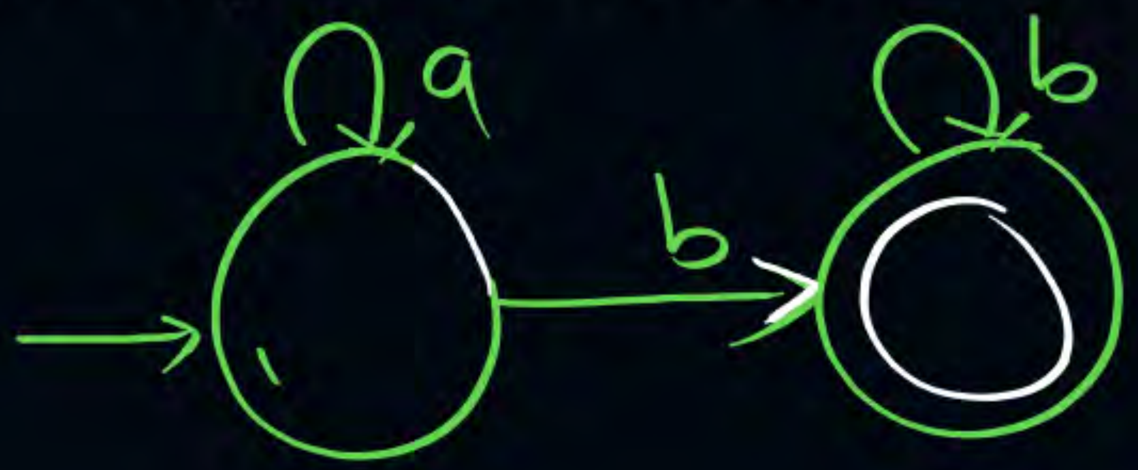


$\Rightarrow a^* + a^*bb^*$

$a^*(\epsilon + b^+) = a^*b^*$

[NAT]

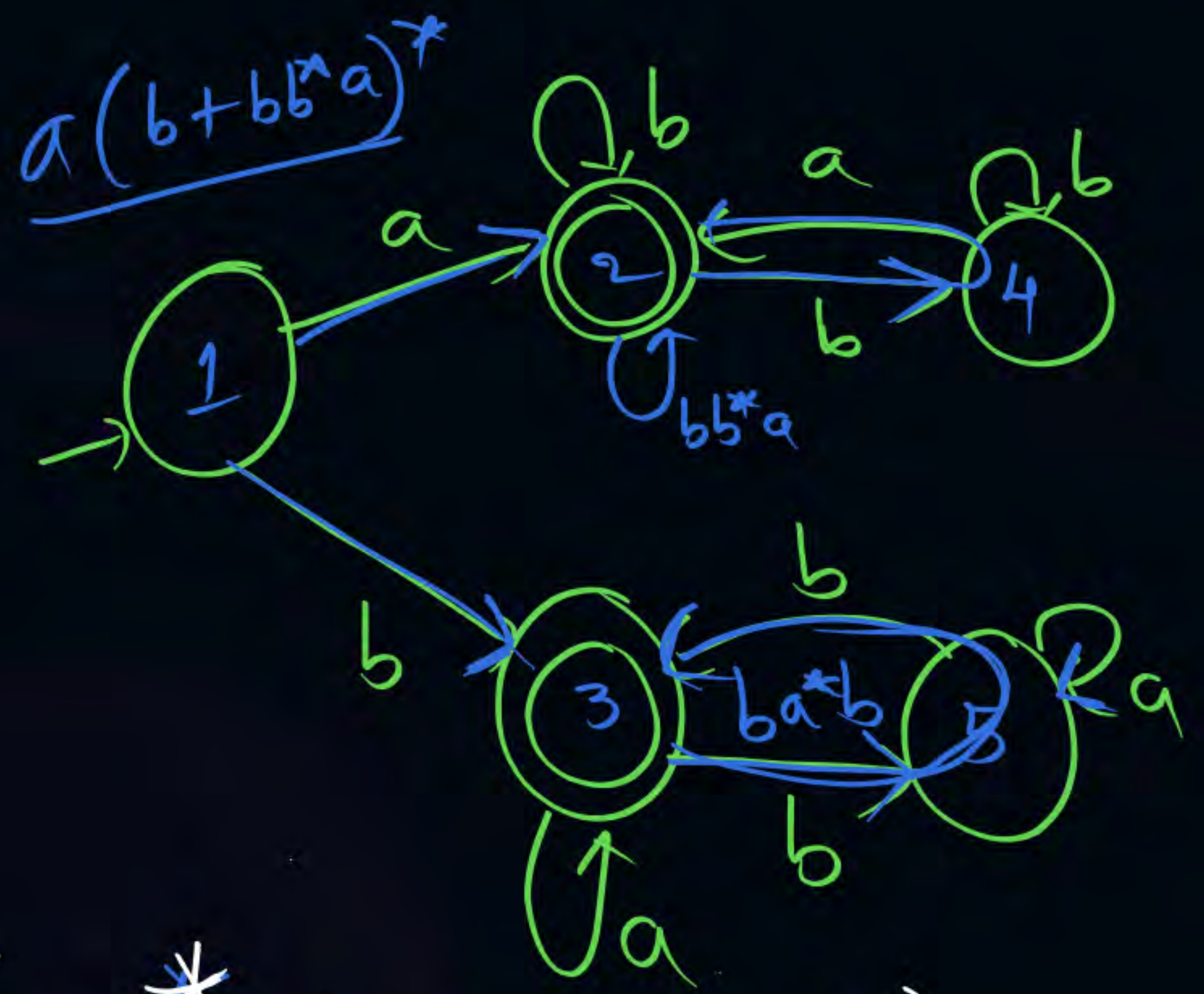
#Q. Construct Regular Expression for the following Finite Automata.



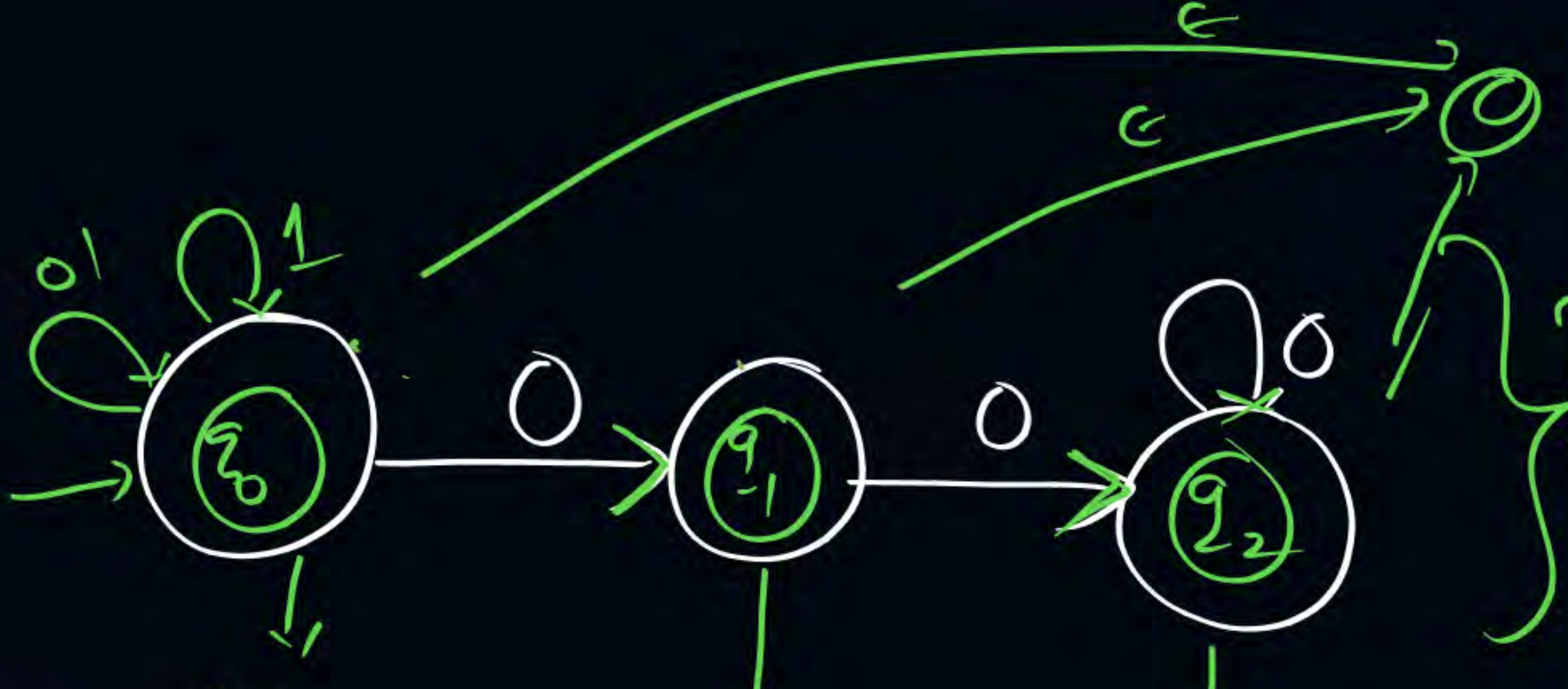
$$\boxed{a^* + a^*bb^*}$$

$$\underline{\underline{a^*}}\,\underline{\underline{(\epsilon + b^+)}} \Rightarrow \boxed{a^*b^*}$$

Regular Expression



$$\left\{ a(b+bb^*a)^* + b(a+ba^*b)^* \right\}$$



Regular Expression

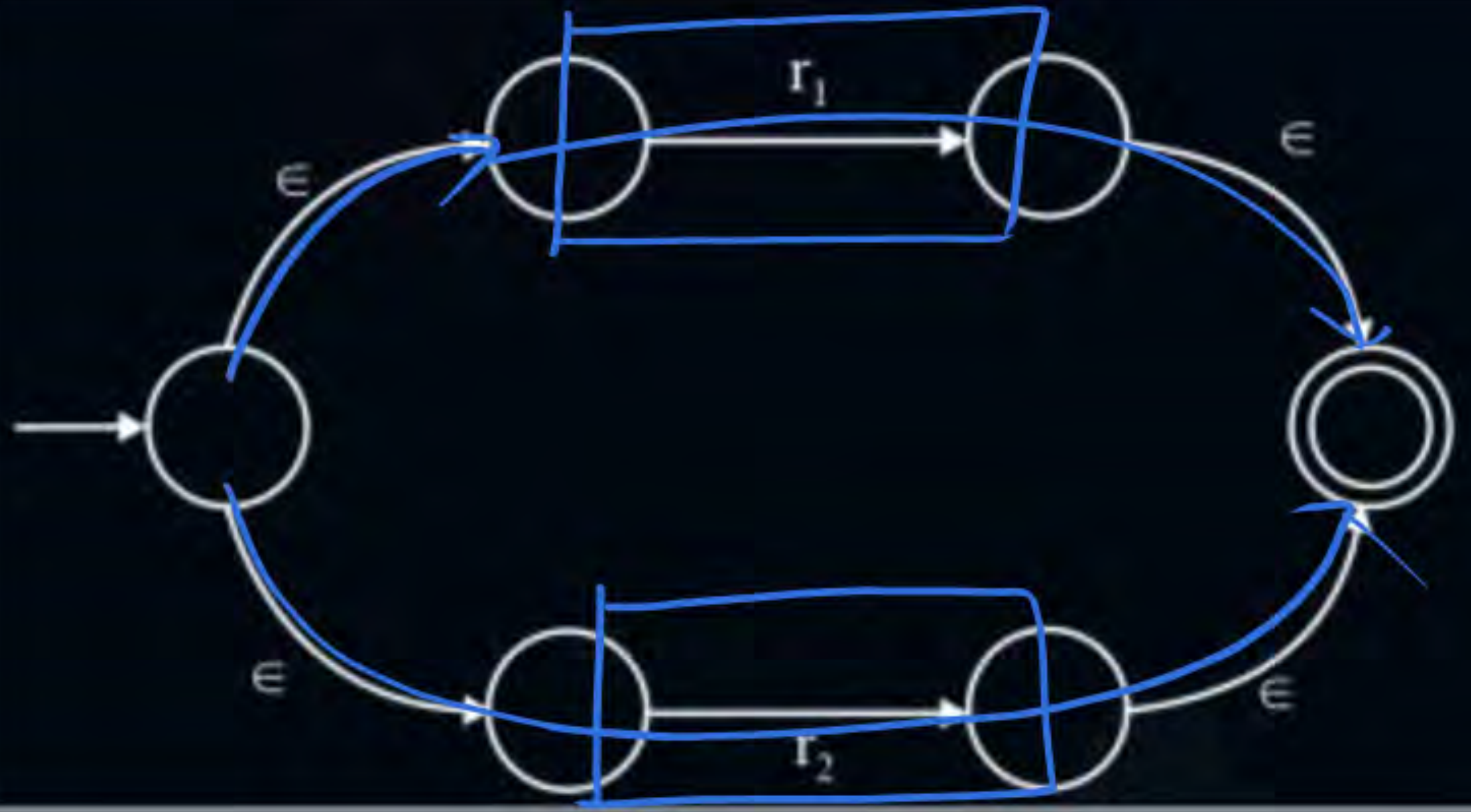
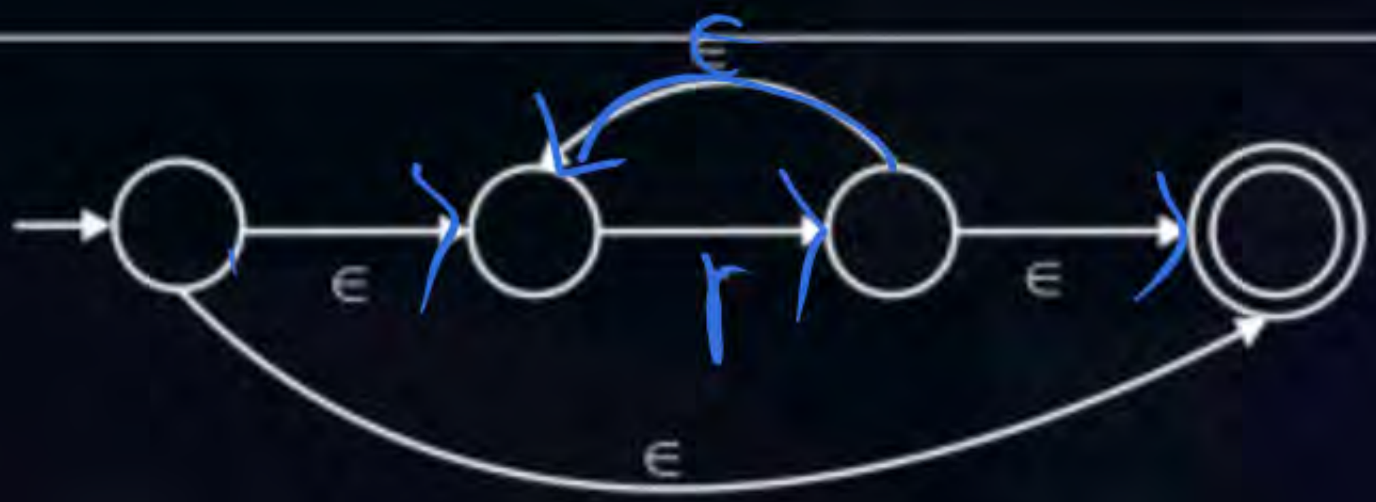
$$(1+01)^* + (1+01)^*0 + (1+01)^*000^*$$ ✓

$$(1+01)^*(\epsilon + 0 + 000^*)$$

$$(1+01)^*0^*$$

Thomson Construction Algorithm

Reg. Expr	∈NFA
1. ϕ	
2. $\in$	
3. a	

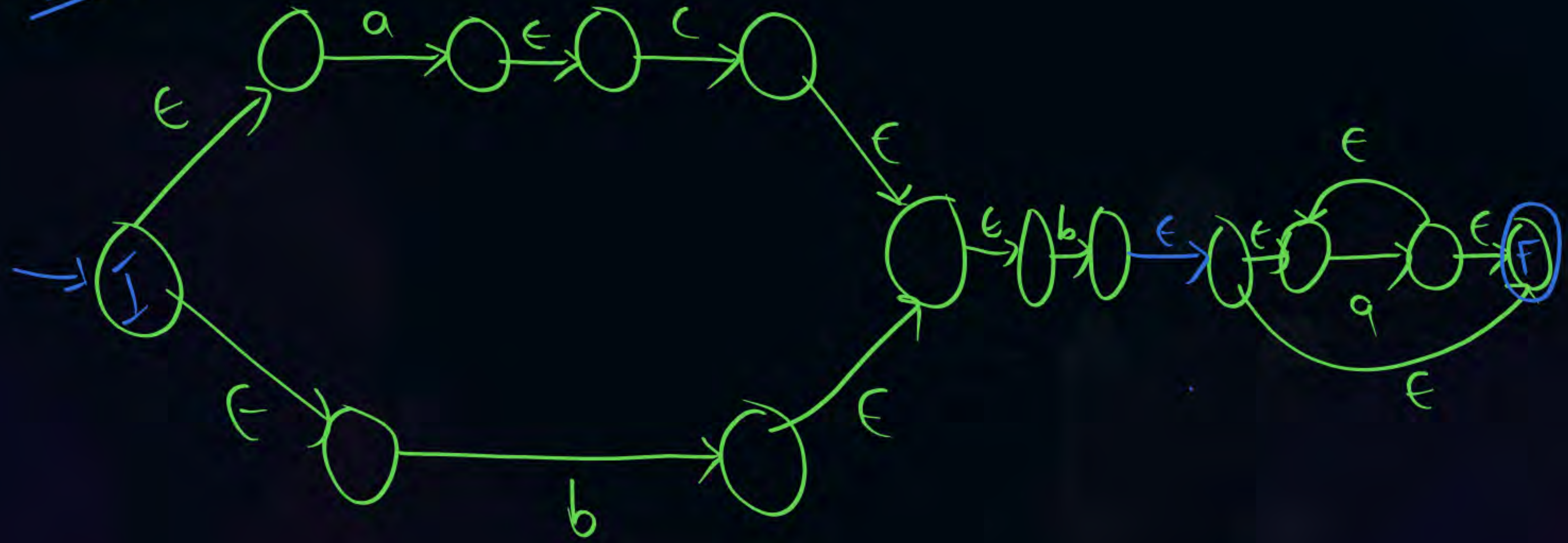
$\{r_1, r_2\}$ 4. $r_1 + r_2$	
5. $r_1 \cdot r_2$	
6. r^* ε	

(1) Construct ε-NFA for given Regular Expression

$(ac+b)\,b\,a^*$

⇓

ε-NFA



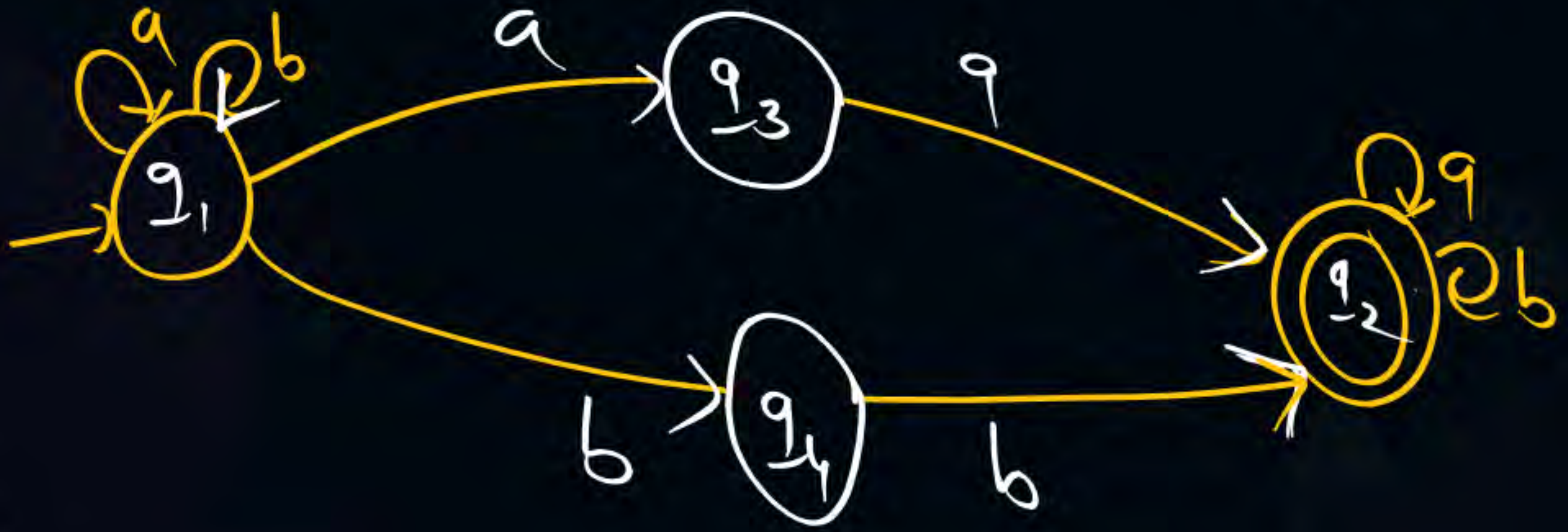
(Q) Construct NFA for given Regular Expression

$$\boxed{(a+b)^*(aa+bb)(a+b)^*}$$

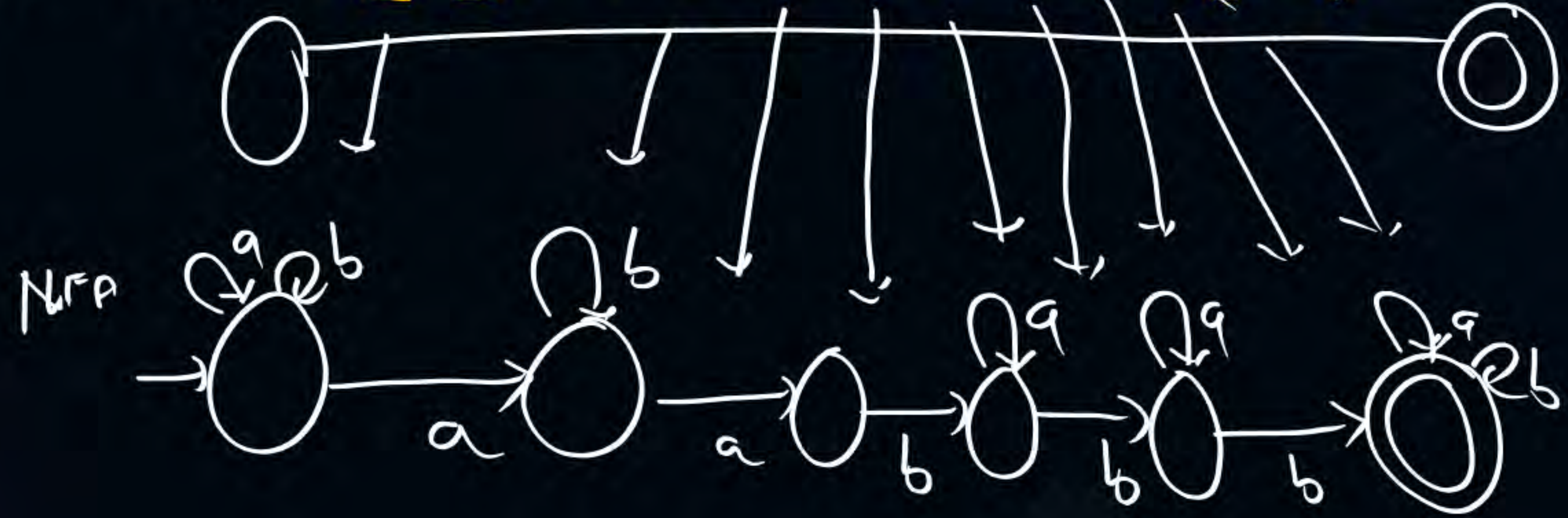
NFA

$\Downarrow$

DFA

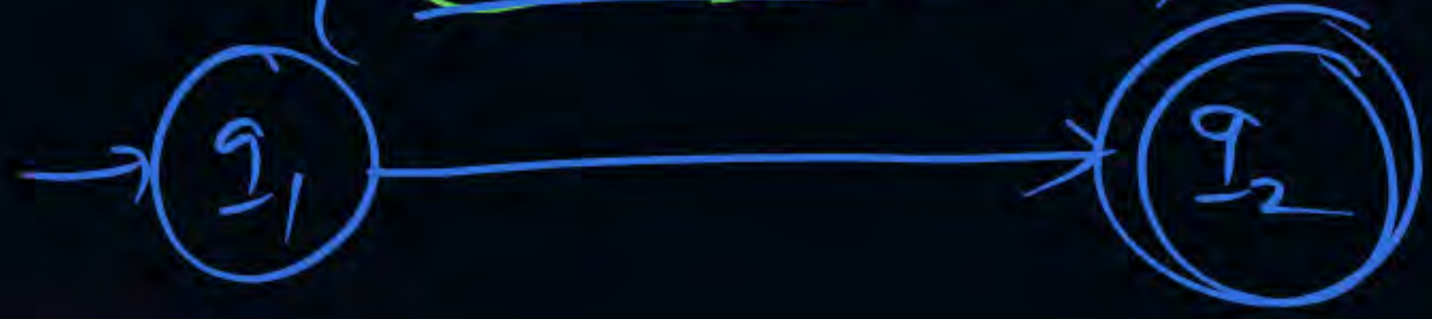


(Q) Construct NFA for $(a+b)^* a b^* a b a^* b a^* b (a+b)^*$



(Q) How many states in min DFA for Regular Expression

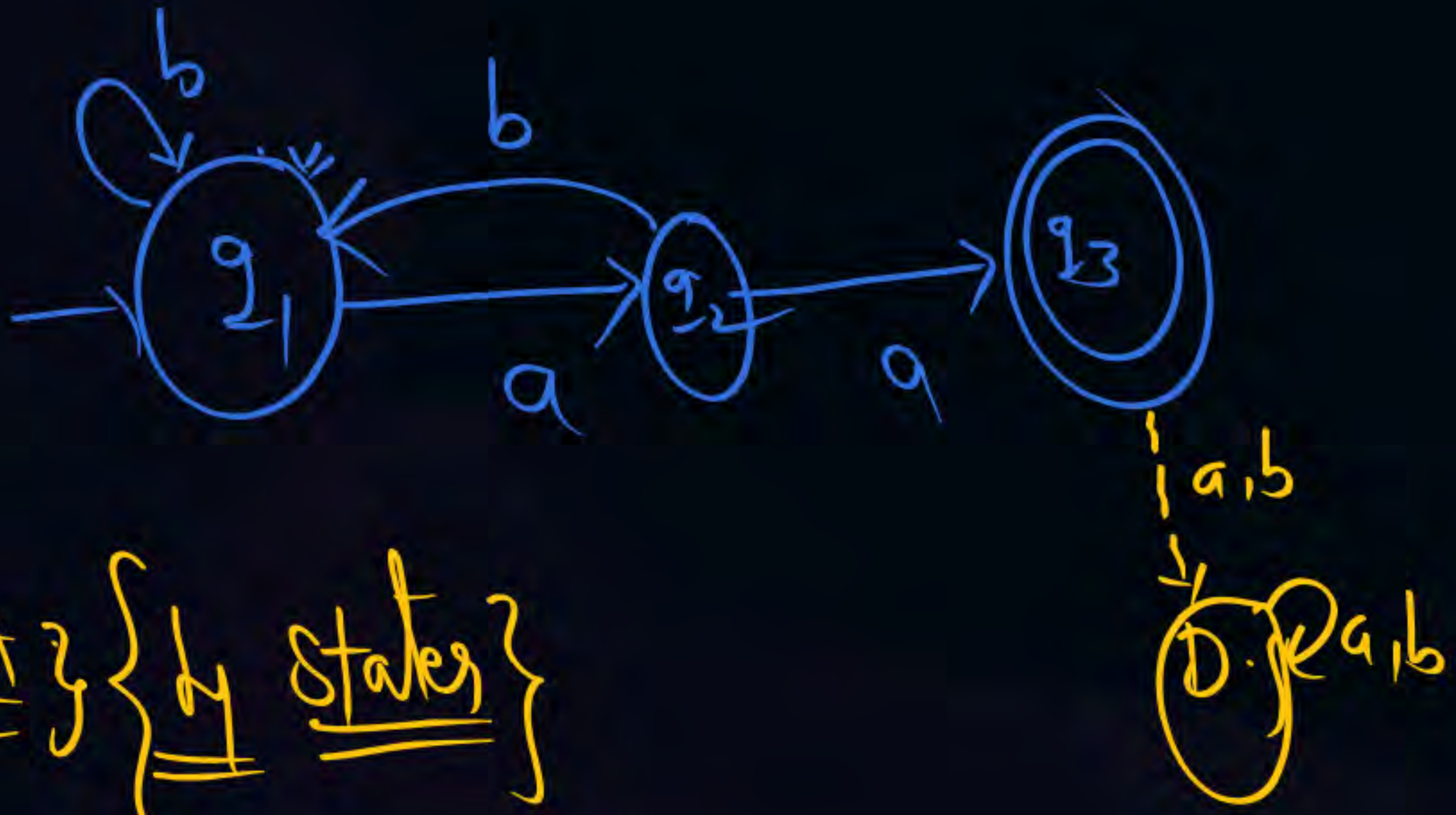
$$\{(ab+b)^* aa\}$$



NFA

$\Downarrow$

DFA } {4 states}



(Q) How many states in min DFA for given Regular Expression

min DFA

1. $(a+b)^4 \xrightarrow{(n+2)}$ 6 States

2. $(a+b+\epsilon)^3 \longrightarrow 5$

3. $[(a+b)^4]^* \longrightarrow 4$

4. $(a+b)^* a (a+b)(a+b) \xrightarrow{2^n} 2^3 = 8$

5. $(a+b)^* abab (a+b)^* \xrightarrow{(n+1)} 5$

6. $b^* a b^* a b^* a b^* a b^* \longrightarrow 6$

min DFA

$a^*b^* \longrightarrow 3$

$a^*b^*c^* \longrightarrow 4$

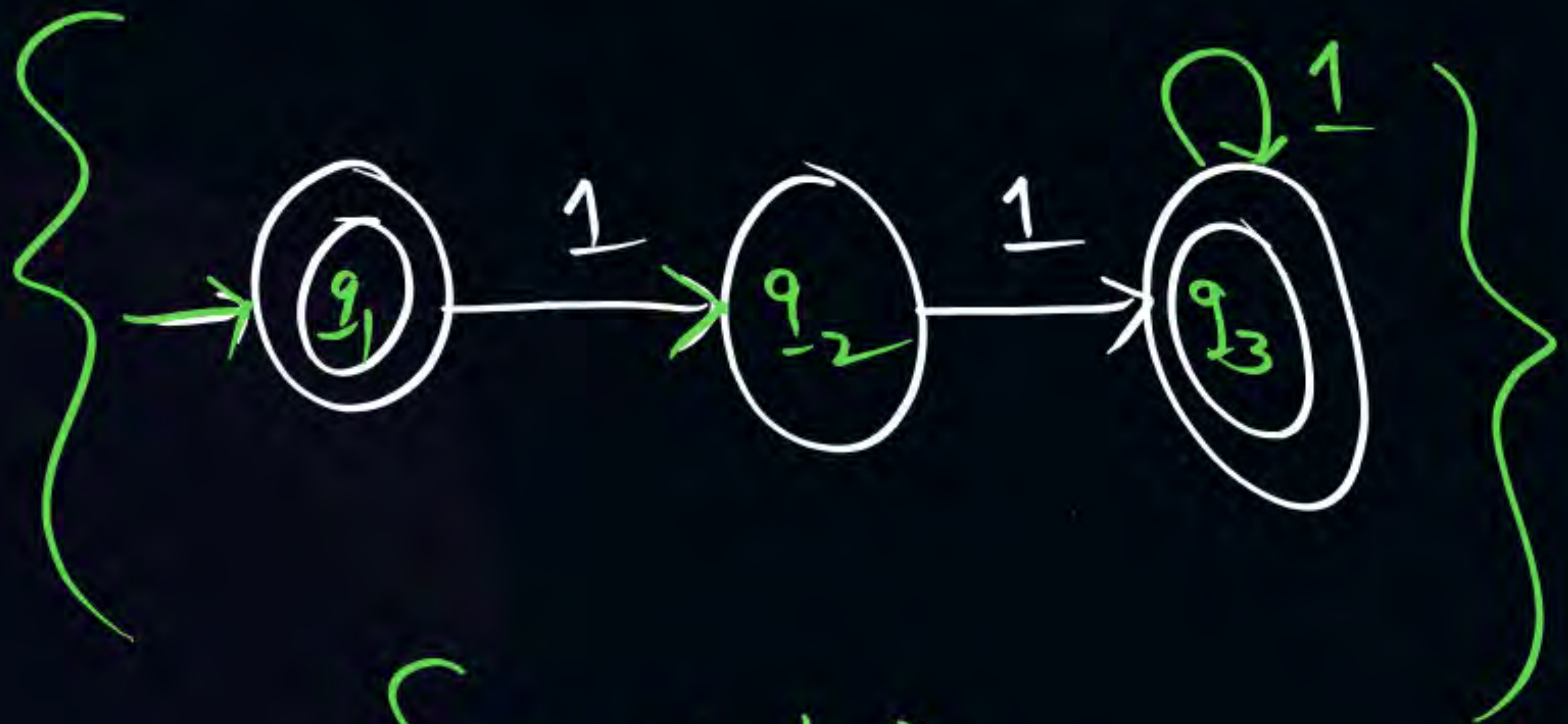
$a^*b^*c^*d^* \longrightarrow 5$

$a^*b^*c^* \ldots z^* \longrightarrow 27$

$0^*1^*2^* \ldots 9^* \longrightarrow 11$

$r = [11 + 111]^*$ } min DFA ?

$\{\epsilon, \underrightarrow{1^2, 1^3, 1^4, 1^5, 1^6, 1^7} \cdots\}$



$\{\underline{\underline{3 \text{ states}}}\}$

THANK - YOU